

Statistical Terms

ANOVA	analysis of variance	dft
ANCOVA	analysis of covariance	dft
CI	confidence interval, e.g., 95% CI	Dft
df	degrees of freedom, see below	OK as is
E or e or <i>e</i>	exponential function	See next page
<i>n</i> , <i>N</i>	number of samples (<i>italic</i>); when <i>n</i> is for individuals, <i>N</i> is for groups. For example, <i>N</i> gives the number of independently performed experiments, and <i>n</i> gives the number of counted/measured events.	OK as is; follow author or query
ND, nd, n.d.	not determined, not defined, not done	dft (query author)
NS, n.s.	not significant (statistically)	dft
OR	odds ratio	dft
<i>r</i>	correlation coefficient	dft
<i>R</i> ²	coefficient of determination	dft
<i>rpb</i>	the point biserial correlation coefficient (when one variable is dichotomous and the other is nondichotomous)	dft
SD	standard deviation	OK as is
SEM	standard error of the mean	OK as is
significant	use only for statistical significance; other meanings should be changed to “substantial,” “important,” etc.	
χ^2 test	chi-square test	OK as is; [spell out without numbers]
<i>z</i> -score	how many SDs an element is from the mean	

Forms for writing statements of probability

<i>t</i> test	$[t_{20} = 2.88, P < 0.01]$	20 is df, pref. subscript (eXtyle)
<i>U</i> test	$(U = 38, n_1 = 19, n_2 = 2, P < 0.01)$	
ANOVA	$[F_{4,30} = 2.26, P < 0.05]$	4,30 are df
<i>F</i> test		
χ^2 test	$\chi^2(7) = 27.13, P < 0.001$ $P = 2.93 \times 10^{-7}$	(7) is df see exponential function below cumulative probability
<i>z</i> -score		

Footnotes in tables or symbols in figures or charts: When probability levels (*P* values) are placed in a table legend, we list them first and use single, double, and triple asterisks; the standard probabilities are below. Remaining footnotes should be placed in the usual order, starting with the dagger; example:

P* < 0.5; *P* < 0.01; ****P* < 0.001. †Indirect immunofluorescence on live cells.

In a figure legend, they appear at the end of a the description of a panel or last in the entire figure description.

Spelling style for probability descriptions

D statistics

Dunnett's test

Fisher's exact test

HSD (honestly significantly different); change to post hoc test

Kruskal-Wallis H statistic

Mann-Whitney U rank sum test

means $\pm$ SEM

Newman-Keuls post hoc comparison

paired t tests with Bonferroni adjustment

NOTE: When the number of contrasts to be estimated is small, (about as many as there are factors)
Bonferroni is better than Scheffé . . ." (www.itl.nist.gov/div898/handbook/prc/section4/prc473.htm)

Scheffé

Student's t test

Tukey or Tukey's [post hoc test]

Wilcoxon rank sum test [for independent samples (W)], signed-rank test [for paired samples (V)]

See, for example, Wikipedia "standard deviation" for graphical illustrations of normal distribution (about three-fourths of the way down the page).

E or e for exponential function in probability statements

Large exponents from au: $p=2.93\text{e-}7$

Change to $P = 2.93 \times 10^{-7}$

from au: $p\text{-value}=6.18\text{E-}09$

$P = 6.18 \times 10^{-9}$

For First Release, follow author (for e or E okay to change to the majority notation)

$P = 2.93\text{e-}7$

or $P = 6.18\text{E-}09$